

Forest Fire Analysis: Predicting Burned Area in Northeast Portugal

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Montesinho Natural Park, Portugal

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Data Dictionary

(Sourced from original dataset, supplemental sources, engineered and transformed features)

- 1. X :** X-axis spatial coordinate within the Montesinho park map: 1 to 9
- 2. Y :** Y-axis spatial coordinate within the Montesinho park map: 2 to 9
- 3. Month :** Month of the year: "jan" to "dec". One-hot encoded and numerically encoded for binary modeling and filtering in the project. ("month_feb, month_num")
- 4. Day :** Day of the week: "mon" to "sun". One-hot encoded and numerically encoded for binary modeling and filtering in the project. ("day_mon, day_num")
- 5. FWI :** Fire Weather Index. Engineered feature.
ISI and BUI scores combined into the ultimate numerical rating to evaluate for general fire intensity potential. Originally created by the National Canadian Forest Service.
- 6. BUI :** Buildup Index. Engineered feature.
DMC and DC combined into a numerical rating to evaluate for the cumulative effects of drought and moisture in relation to the FWI System overall. It responds to fuel accumulation as well.
- 7. FFMCI :** index from the FWI system: 18.7 to 96.20
Previous FFMCI, temp, RH, wind speed and rain combined into a numerical rating of the moisture content of litter and other cured fine fuels. Used to represent the potential for human-caused ignition. It responds to temperature, relative humidity, wind and rain, and indicates the relative ease of ignition and flammability of fine fuels.
- 8. DMC :** index from the FWI system: 1.1 to 291.3
Previous DMC, temp, RH, rain and month of the year combined into a numerical rating of the average moisture content of loosely compacted organic layers of moderate depth. Used to assess the potential for lightning-caused ignition. It responds to temperature, relative humidity and rain, and provides an indication of fuel consumption in moderate duff layers and medium-sized woody material.
- 9. DC :** index from the FWI system: 7.9 to 860.6
Previous DC, temp, rain and month of the year combined into a numerical rating of the average moisture content of deep, compact organic layers. It responds to temperature and rain and indicates the effects of seasonal drought on forest fuels and the potential for smouldering in deep duff layers and large logs.
- 10. ISI :** index from the FWI system: 0.0 to 56.10
FFMC and wind speed combined into a numerical rating of the expected rate of fire spread. Higher values indicate faster spread rates. It combines wind speed with the FFMCI without accounting for the quantity of fuel available for combustion.
- 11. Temp :** Temperature in Celsius degrees: 2.2 to 33.30
- 12. RH :** Relative humidity in %: 15.0 to 100
- 13. Wind :** Wind speed in km/h: 0.40 to 9.40
- 14. Rain :** Outside rain in mm/m2 : 0.0 to 6.4
- 15. Area :** The burned area of the forest (in ha): 0.00 to 1090.84
- 16. Log_area :** Engineered feature. Original 'Area' feature was extremely skewed towards 0.00, thus leading to logarithmic transformation for better modeling and evaluation.

1. Detailed EDA & Preprocessing Decisions

With three team members, this project began with independent exploratory data analyses conducted by each member. This decision was made as a group to gather many various findings and after discussion, compile the most valuable findings into one singular plan of action for preprocessing. The dataset was sourced from the raw UCI Montesinho Natural Park forest fires dataset. All three analyses converged on similar core findings, which gave our team confidence that the patterns identified were genuine and verifiable. This collaborative approach continued throughout the entire process.

1.1 Data Quality

- The initial dataset consisted of 517 rows and 13 columns.
- No missing values were found present anywhere in the dataset.
- 4 exact duplicate rows were identified in the raw data. These were removed by Lofinda during her own preprocessing stage, but ultimately left in the consolidated dataset used for supervised learning. Amir's clustering and supervised modeling pipeline used the full 517 rows in addition to his engineered features and remained consistent in his analysis.
- The target variable 'area' was extremely right-skewed to equal 0.00.
- A small number of outlier fires burned several hundred to over 1,000 hectares.

1.2 Target Transformation

Because of this skew, $\log_area = \log(area + 1)$ was created and used as the main regression target for both visualizations and modeling. The $\log_{1p}$ form was chosen specifically to handle the large number of zero-area observations, since $\log(0)$ is undefined. This transformation was also recommended by the dataset's original authors (Cortez & Morais, 2007).

1.3 Feature Engineering

Amir engineered BUI (Buildup Index) and FWI (Fire Weather Index) from the existing composite columns by following the official Canadian Forest Fire Weather Index System's formulas. These composite indices were not present in the original dataset and became a central point for his modeling processes tested and built down the line.

'Month' and 'day' were also encoded in two ways for separate downstream use: ordinal numeric encoding (month_num, day_num) for use in regression/tree models by Amir, and one-hot encoding for exploratory grouping and filtering by Lofinda.

1.4 Supplemental Visual Findings

Beyond the target transformation, each group member's EDA produced a set of exploratory histograms and scatterplots examining the raw weather variables and fire weather indices individually. While these did not all lead to further dedicated analysis given the project's time constraints, several patterns were worth noting and illustrated in 1.5 EDA Figures.

Fire observations were heavily concentrated in August and September (**Figure 1**). While August recorded more fires than September, the fires in September produced a greater total burned area, highlighting how frequency and severity did not track perfectly together.

The extreme skew of the target variable (area) is visible directly in its distribution (**Figure 2**). There is almost no visible spread beyond the zero mark. Applying the log transformation does reveal more spread and a lower peak, confirming that this transformation was necessary for meaningful analysis and model building.

Histograms of the relationship between frequency and the feature variables (**Figure 3**) shown reflect the fire-friendly conditions when these fires occurred. FFMFC, DC and BUI skew heavily toward high-risk ranges, while temp shows the only relatively normal distribution centered around 18-22 °C. Rain shows almost no variation. Overall the weather variables reflect logical and expected behavior.

Early scatterplots of the raw FWI indices against $\text{Log}(\text{Area}+1)$ (**Figure 4**) indicate that FFMFC was the only variable demonstrating a weak relationship to fire size. DMC, DC and ISI showed no real patterns, with large and small fires recorded across the range.

A separate pair of scatterplots (**Figure 5**) examined how the overall FWI related to two of it's combined inputs, temperature and wind. Temperature showed a clear, positive relationship with FWI, while wind showed no apparent relations. This was an early signal that some raw weather variables were more valuable than others for modeling purposes.

The correlation matrix (**Figure 6**) included the raw weather variables and composite indices, as well as the engineered composite features built by Amir, and displays their relationships to burned area. None of the variables showed a strong correlation with area. The highest score was temperature at only 0.10. This foreshadowed a reoccurring theme of our analysis: no matter how composited, the individual variables offered weak support for predicting burned area.

1.5 EDA Figures

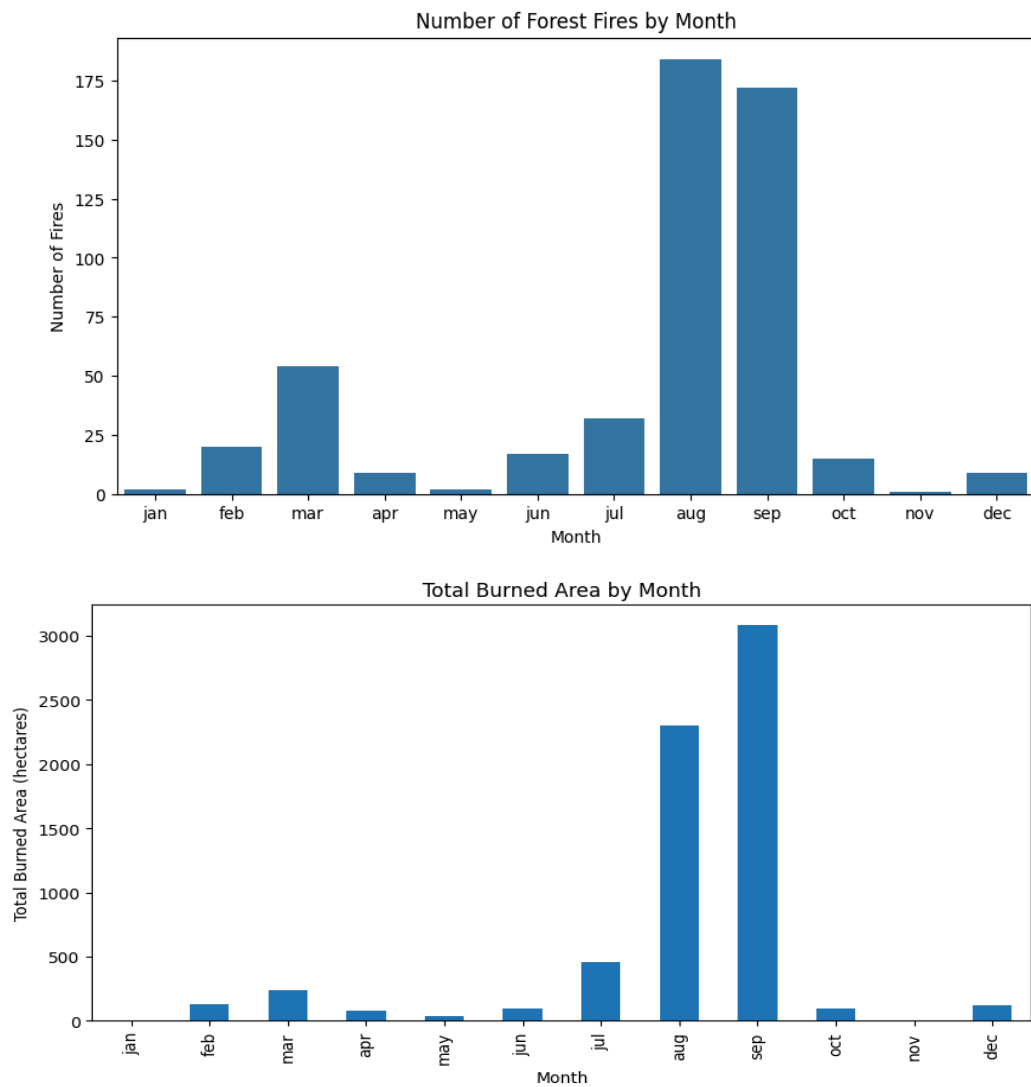


Figure 1.1 Number of forest fires by month (top) and total burned area by month (bottom).

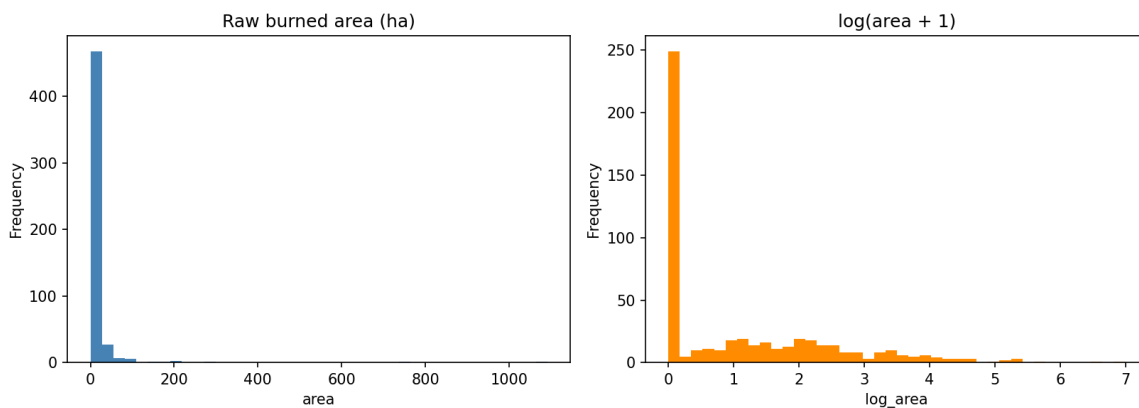


Figure 1.2 Raw burned area vs. Frequency (left) and $\log(\text{area} + 1)$ vs. Frequency (right).

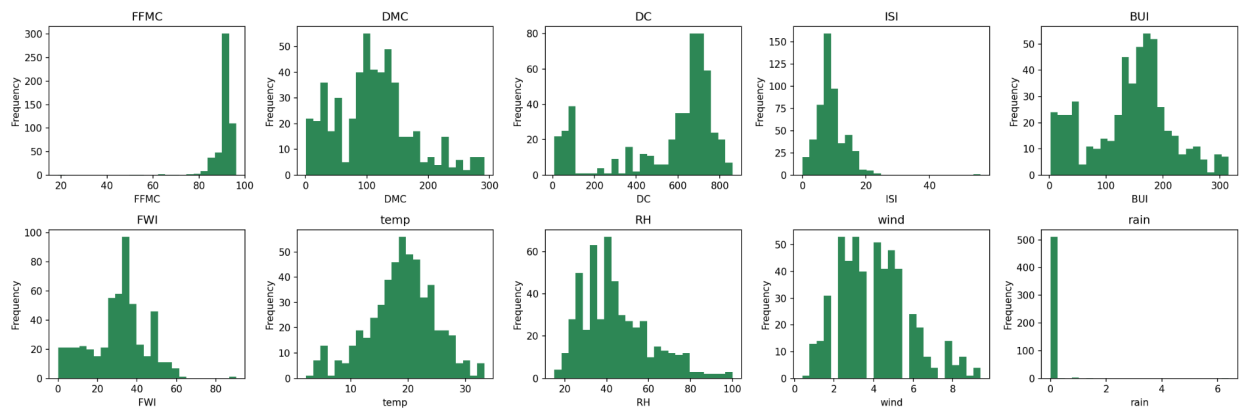


Figure 1.3 Distributions of FFM, DMC, DC, ISI, BUI, FWI, temperature, relative humidity, wind, and rain. (from left to right).

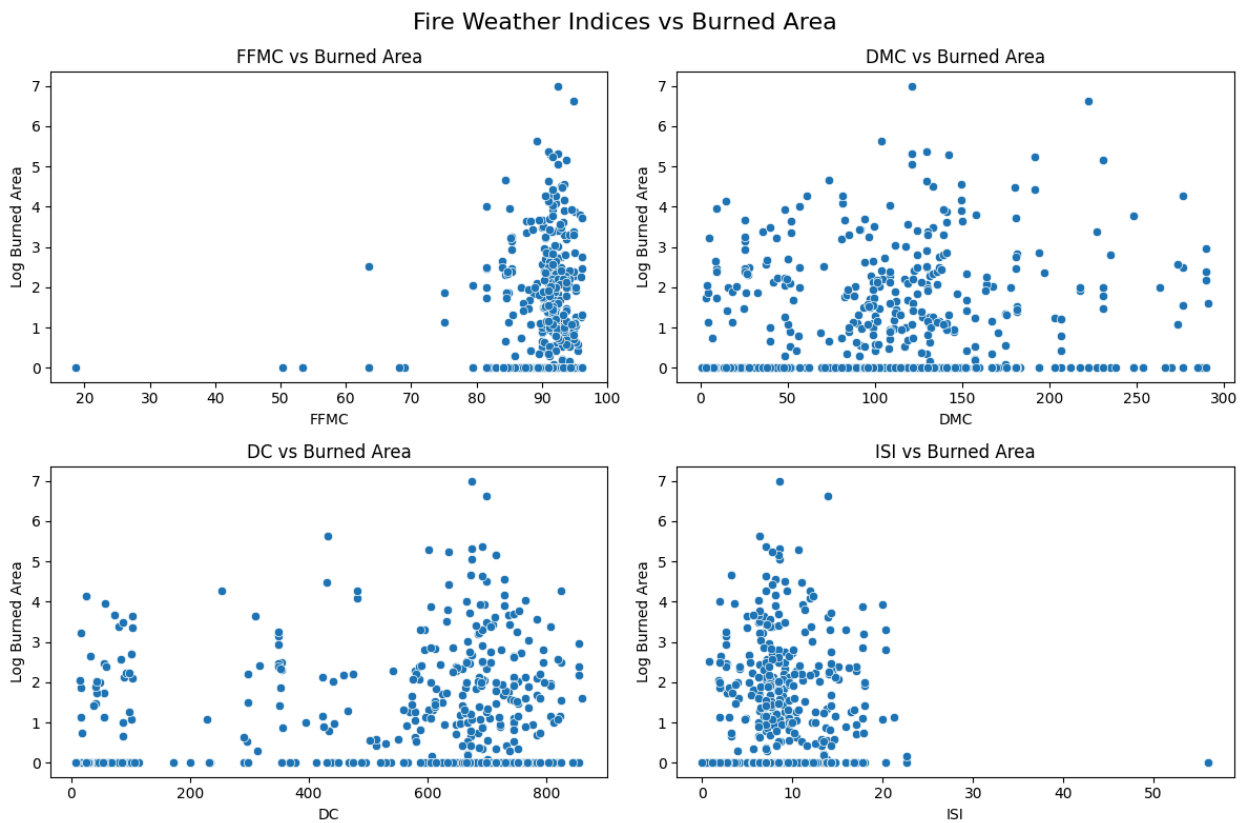


Figure 1.4 FFM, DMC, DC, and ISI vs. Log(Burned Area+1).

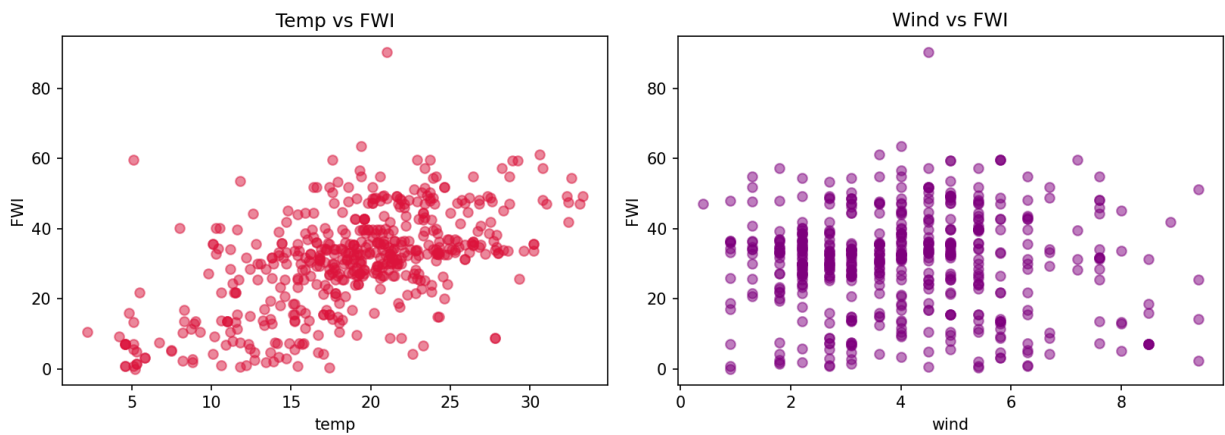


Figure 1.5 Temperature vs. FWI (left) and wind vs. FWI (right).

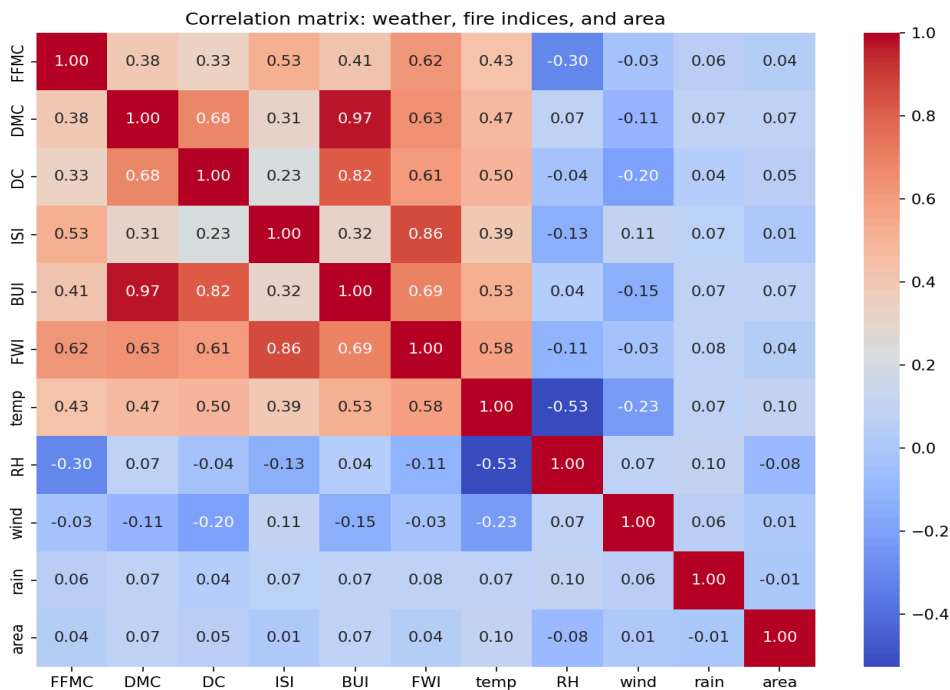


Figure 1.6 Correlation matrix across weather variables, fire-weather indices, and burned area.

1.6 Collaborative Plan of Action

After our group compared initial EDA findings, we split the remaining work into roles. Amir and Lofinda continued with their own preprocessing and unsupervised learning, with Amir being the sole member to continue with supervised learning. I took on the role of team lead, holding myself responsible for guiding and consolidating our team's work, building the final Tableau dashboard, and authoring this report. Lofinda created the Github repository and was responsible for its maintenance. Lofinda also built a Tableau dashboard and after group review, mine was selected as the version for submission.

2. Unsupervised Learning Approach & Findings

To help prevent bias, both Amir and Lofinda independently applied K-Means clustering to test whether fire-weather conditions naturally separated into distinct groups, without using burned area to define the clusters. The two approaches used different feature sets and arrived at a different number of clusters, but reached otherwise compatible conclusions.

2.1 Lofinda's Approach ($K = 2$)

Lofinda clustered on raw meteorological data and fire-risk variables (temperature, wind, humidity, and fire-risk indicators). The elbow method and silhouette score supported a two-cluster solution. The resulting groups separated into a hotter/drier cluster and a cooler/higher-humidity cluster.

Key finding: The hotter/drier cluster contained 77% of all fires and had a substantially higher average burned area (14.97 ha) than the cooler/more humid cluster (5.94 ha). Lofinda concluded that hotter, drier conditions tend to be associated with larger fires, but that the two groups still had considerable overlap. In short, weather conditions alone do not fully explain fire size, and some other unmeasured factors likely play a role.

2.2 Amir's Approach ($K = 3$)

Amir clustered on a six feature set chosen to balance four categories of information given in the dataset: surface fuel state (FFMC), fuel accumulation (BUI), composited Fire Weather Index (FWI), and raw weather (temp, wind, RH). This differed from Lofinda's approach in that he substituted his engineered composite indices (BUI, FWI) for the raw underlying composites (DMC, DC, ISI). Amir's elbow method bent somewhere between $K=2$ and $K=4$, leading him to select $K=3$ as the middle point.

This resulted in 3 distinct clusters:

Cluster	n	Mean Area (ha)	Median Area (ha)	Avg FFMC	Avg BUI	Avg FWI	Avg Temp (°C)
0 — Low Risk	114	6.08	0.00	85.23	37.42	10.36	12.51
1 — Hot/Dry High Risk	300	17.07	1.04	92.47	167.80	37.13	22.29
2 — High Fuel, Humid, Contained	103	8.05	0.00	91.33	183.58	34.33	16.04

2.3 Unsupervised Visual Figures

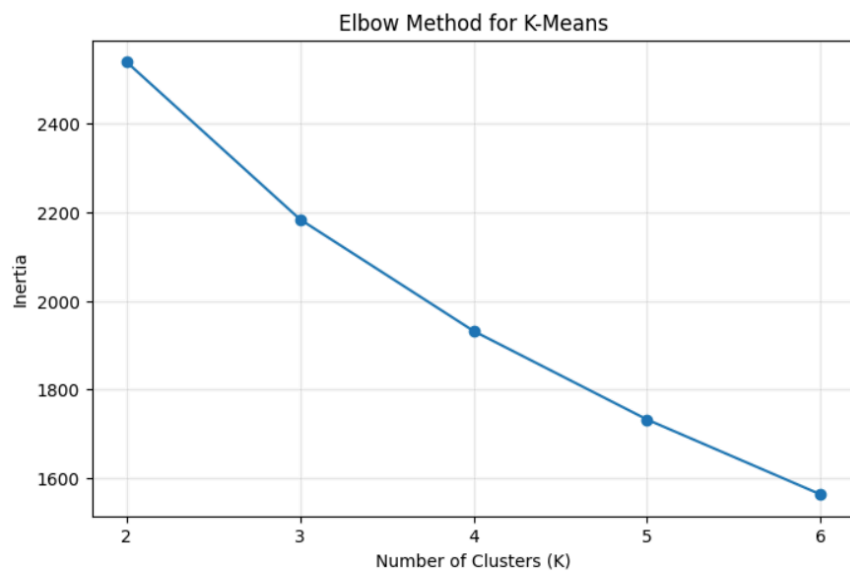


Figure 2.1 Lofinda's Elbow Method for K-Means results.

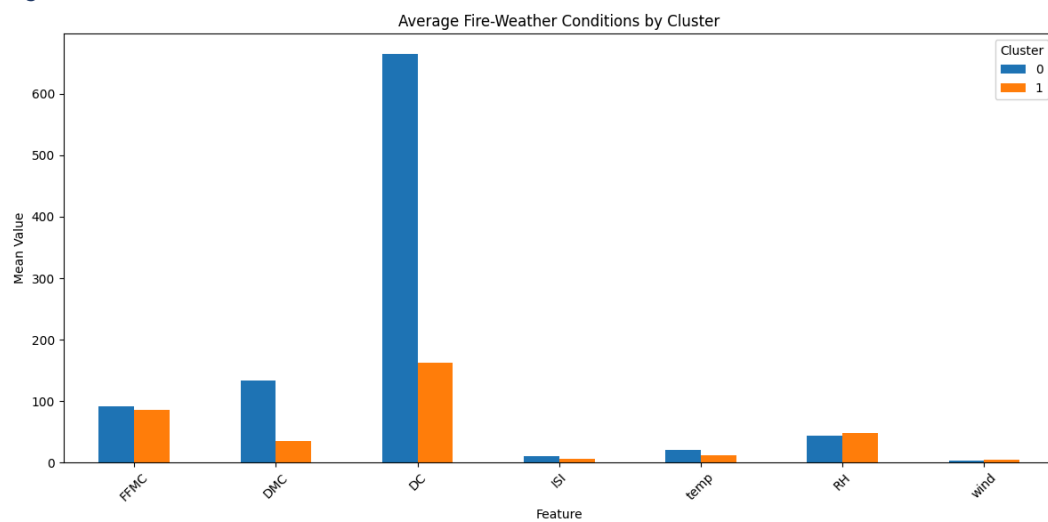


Figure 2.2 Lofinda's K=2 Clusters charted by Average Fire-Weather Conditions.

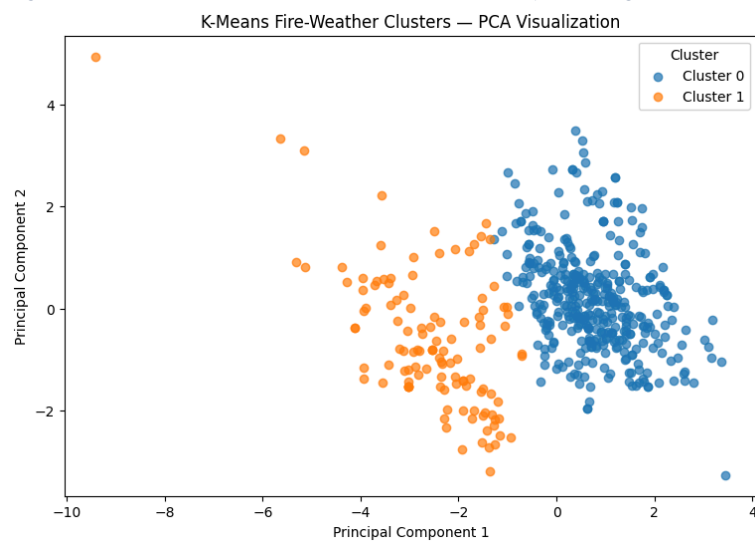


Figure 2.3 Lofinda's K=2 Clusters PCA Visualization.

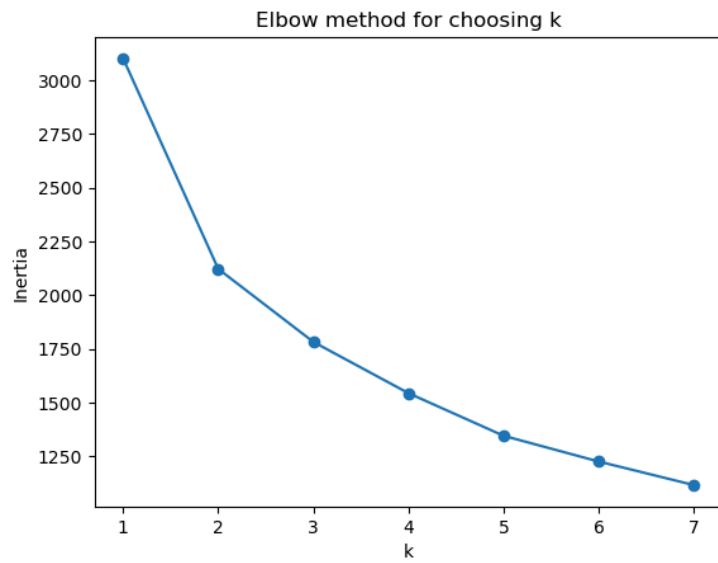


Figure 2.4 Amir's Elbow Method for K-Means Results.

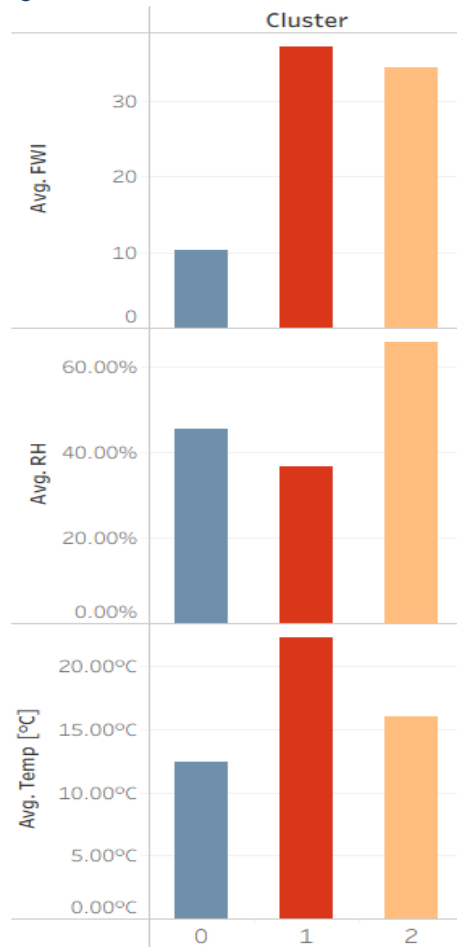


Figure 2.5 Amir's K=3 Clusters charted by Average FWI, RH and Temp °C.

2.4 Benefits of Two Unsupervised Model Approaches

Both Lofinda and Amir used different feature sets (raw components vs. engineered composites) and different scaling/elbow interpretations, which organically produced different elbow curves and cluster counts. Both were legitimate and well-justified in the choices they made while constructing their pipelines. Since their findings reached a convergent conclusion - that hot, dry, high-fuel conditions were associated with more fire spread, but not enough for determination - our group viewed this as a strength rather than a weakness.

3. Supervised Models Tested & Compared

Motivated by his prior experience as a math teacher with regression, and a determination to improve upon the model found in the source literature, Amir led the supervised learning portion. Amir had a goal of building a model that would improve meaningfully on the previous correlations. Eight model configurations were tested in total, moving from simple interpretable baselines towards more progressively informed feature sets.

3.1 Amir's Ambitious Approach

Models A and B (Linear Regression) served as the baselines, assuming linear relationships which fire risk and behavior likely violated. Models C, D and E (RandomForest) tested whether capturing nonlinearity and feature interactions could improve upon the baseline. fwi_compile (RandomForest) tested the full FWI System chain with no raw weather at all. Model F (RandomForest) further tested by adding wind and rain on top of the best base model found so far, and Model G tested adding the unsupervised cluster feature on top of the same base.

Model	Features	Type	MAE (area)	RMSE (log)	R ²
A	FWI, temp, FFMC, RH	Linear Regression	19.83	1.48	-0.001
B	FWI, FFMC, RH, wind	Linear Regression	19.81	1.48	0.011
C	FFMC, DMC, DC, temp	Random Forest	19.68	1.49	-0.020
D	temp, ISI, BUI, FWI	Random Forest	19.98	1.52	-0.054
E	FFMC, DMC, DC, ISI, BUI, FWI, temp	Random Forest	19.75	1.49	-0.018
fwi_compile	FFMC, DMC, DC, ISI, BUI, FWI (no raw weather)	Random Forest	19.45	1.45	0.041
F	fwi_compile features + wind + rain	Random Forest	19.75	1.46	0.024
G	fwi_compile features + cluster	Random Forest	19.42	1.44	0.057

3.2 Notable Patterns Across Regression Models

- All three RandomForest variants built from single/overlapping raw variables (Models C, D, E) produced negative R² scores, underperforming even the weak but positive Model B (0.011). Increasing model flexibility did not improve predictive performance. If anything, it hurt performance, suggesting that the limitation here was caused by the dataset rather than any model complexity.

- Fwi_compile (full FWI chain, no raw weather) outperformed every model built specifically for the redesign (C, D, E), despite being a simpler feature set. This was the first positive R^2 during the supervised modeling sequence.
- Adding raw weather back on top of 'fwi_compile' consistently hurt performance: Adding temp alone (Model E) dropped R^2 from 0.041 to -0.018; adding wind and rain (Model F) dropped it to 0.024. This is one of the more consistent and interesting findings throughout our analysis. The FWI system's indices appear to already carry whatever weather signal is usable, and raw weather variables layered on top mostly just added noise.
- Model D (composite indices and temp) was the weakest of the RandomForest variants. Despite having four named features, this set offered few genuinely independent directions for the model to split on. With a training set of only 413 records, this redundancy likely made the model more prone to overfitted noise rather than learning structure.

4. Model Selection & Rationale

Model G (fwi_compile feature set combined with unsupervised cluster label) was selected as the final model. It was the best performing model across the entire testing sequence (A-G), and the only weather-derived addition to the analysis that improved rather than degraded predictive performance.

4.1 Why Model G Stood Out

Unlike every previous model, which tested every raw weather variable in addition to the FWI indices, Model G improved every metric over the base model by adding the cluster label ($MAE(\text{area}) = 19.42$, $RMSE(\log) = 1.44$, $R^2 = 0.057$). This supported the central hypothesis behind the clustering analysis: individual raw weather variables carried too little independent signal, and tended to introduce overfitted noise, but the same underlying conditions compressed into a single risk category via unsupervised clustering, captured a combined multivariate structure that no single raw variable could express on its own. The improvement from fwi_compile ($R^2 = 0.041$) to Model G ($R^2 = 0.057$) was modest at best, and should be cautiously interpreted given the limited test size ($n = 104$). Still, it was the first and only analysis that improved upon predictive performance, which is why we chose it as our final model.

5. Detailed Results

5.1 Predicted vs. Actual (Model G)

Most points in the predicted vs actual plot sat well off the perfect-prediction line, especially the handful of large actual fires ($\log(\text{area}+1)$ above 4), which the model consistently underpredicted. The largest fire in the test set (actual $\log(\text{area}+1) \approx 7.0$) is predicted at only ≈ 3.3 , and a second large fire (actual ≈ 5.3) is predicted at only ≈ 2.1 . There is also a dense cluster of points at Actual = 0, with predicted values ranging from 0 up to ≈ 2.6 , highlighting how the model remained noisy even on the fires with no measurable spread.

Even with the clusters added, Model g's core weakness persists. It cannot reliably predict large fires.

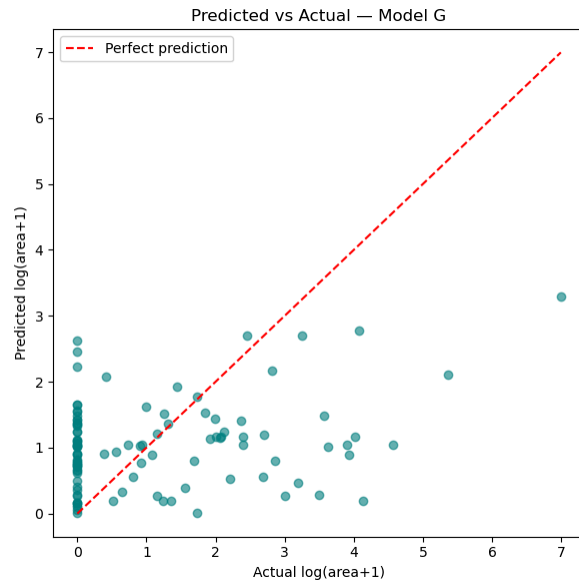


Figure 5.1 Predicted vs. Actual Log(Area+1) - Model G.

5.2 Overall Metric Summary

Across all eight models tested (A-G plus the fwi_compile baseline), R^2 values ranged from -0.054 to 0.057. Even the best model could only explain about 5-6% of the variance in burned area. This was consistent with the weak univariate correlations observed in the heatmap (side: strongest single-variable correlation was 'temp' at 0.10), and reflected the well documented difficulty of this dataset from the original source.

6. Dashboard Interpretation

Both Lofinda and I independently built our own Tableau dashboards to communicate our group's findings to a non-technical audience. Lofinda's dashboard effectively highlighted key metrics and earlier analysis points, but didn't include the supervised learning as part of the narrative. After group review, and keeping presentation in mind, our group selected my own dashboard as the version used for submission and presentation. Despite this being my first time using Tableau, I was able to effectively tell a cohesive story across our analytical journey.

The selected dashboard included 8 linked visuals that mirror the structure in this report:

- **Fires by Month / Burned Area by Month:** Established the concentration of fire activity in August and September, and the distinction between fire frequency and fire severity.
- **Correlation Heatmap:** Visualized the core challenge of the project: no individual weather variable or fire index correlated strongly with burned area.
- **FFMC vs. Log(Area+1):** Highlighted the one raw variable that showed a real (nonlinear, threshold-like) pattern with fire size, and noted that FFMC is itself a composite of prior FFMC, temperature, humidity, wind, and rain — motivating Amir's later engineering of BUI and FWI.

- **One Cluster Stood Alone / K-3 Clusters:** Amir's clustering results, showing that Cluster 1 is the only group where measurable fire spread is the typical outcome, and that humidity — not fuel or danger scores alone — differentiated Cluster 1 from Cluster 2.
- **Eight Models, One Modest Win: Model G:** Summarized the full A–G model spectrum and highlighted Model G's modest but real improvement.
- **Predicted vs. Actual: Model G:** Completed the dashboard with an honest reality check on the limits of the best model produced.

Note: Since Amir was the sole member of our group able to deliver the presentation in person, I moved around the order of the visuals into a structure that he felt most comfortable presenting the dashboard and telling the story of our process. Both versions will be listed in the 'Forest_Fire_Analysis' Repo, filed under /Dashboards.

7. Limitations

Our group analysis confirmed what the dataset's original authors (Cortez & Morais, 2007) and subsequent published works have consistently found: predicting burned area from meteorological and fire-index data alone is an incredibly difficult regression problem. Across every model configuration tested (linear and nonlinear, raw and composite features, with and without K clusters), no model could strongly predict burned areas for Montesinho National Park.

- **Weak underlying signals:** No single raw weather variable or FWI indice correlated strongly with burned area. The strongest correlation was temp with only 0.10.
- **Small dataset:** The raw dataset held less than 1,000 rows (518). After preprocessing there were only 513 usable observations, with approximately 400 used for training and 100 saved for testing after the train/test split. This limited the statistical power available for the final model (G) to only predict with mild success.
- **Extreme skew of target variable:** Nearly half of all the observations had an area=0, with the largest fires being outliers among the dataset. Even after compensating with Log(Area+1), every model tested vastly underpredicted the large fires.
- **Feature redundancy across composite indices:** FWI, BUI, FFMC, and many more indices were features used in this project. Because of their over-lapping nature, this reduced their usefulness in tree-based model building and instead led to increasing noise and potential overfitting.
- **Missing possible additional variables:** The original dataset did not include any other meaningful factors related to forest fire risk, such as: fuel types, ignition cause, terrain, fire fighters response time, or more. Any supplemental data could have changed the direction of this project and potentially improved analysis or prediction.

7.1 Time - Structural Limitations Examined

Taking a closer look at the limitations of this dataset, time was an underutilized factor collected for further analysis. The original Montesinho dataset observations only include records for month and day of each fire, with weather and fire risk indices representing the daily conditions. In a recent publication

from the Canadian Forest Fire Danger Rating System (FWI25-26) , the update explicitly recommends moving from once-per-day weather observations to hourly weather inputs, noting that this allows fire agencies to “better account for changes throughout the daily weather cycle” (Natural Resources Canada, 2025, Section 4.4&5). The same publication acknowledges that while hourly data increases analytical complexity, it also enables more accurate fire potential assessments than daily and monthly data allows.

This directly ties back to our current predicament: the raw weather and composite FWI indices used throughout this project vary meaningfully within a single day. Temperature, humidity and wind in particular can have variations that swing across the spectrum of interpretation in a rapid hourly fashion. It is entirely plausible that some of the weak correlations and modest model performance could be remedied if we had access to a more rich and complex amount of hourly data over daily data. Hourly data would also include an overall larger dataset. More data could lead to better testing and stronger models, thus leading to better evaluation scores and clearer prediction.

8. Recommendations

- **Acquire hourly-resolution weather and fire index data.** Per Section 7.1, the current national guidance for the FWI system itself has moved towards adopting hourly inputs. In addition, the FWI2025 now includes support and guidance on monitoring sunrise, sunset and sunlight hours, as well as peak burn hours. “While hourly outputs may be an overwhelming amount of data from the daily regional planning perspective, this real-time monitoring can highlight discrepancies and potential shifts in fire behavior...this functionality could provide early indications of unexpected activity...” (Natural Resources Canada, 2025, Section 4.4&5).
- **Incorporate missing contextual variables not currently present in the dataset.** Terrain/elevation, vegetation and fuel types, ignition cause, fire watch response time, and more could help bring more clarity or correlation to a small and weak dataset.
- **Treat forest fire potential and fire size as two separate modeling problems to target.** Given that roughly half of the data had zero burned area, perhaps a two - stage approach could provide a better structure for interpreting and assessing the data at hand. For example: classification followed by regression, rather than solely regression.

Sources

UCI Machine Learning Repository — Forest Fires Dataset: <https://archive.ics.uci.edu/dataset/162/forest+fires>

Cortez, P., & Morais, A. (2007). A Data Mining Approach to Predict Forest Fires using Meteorological Data.

Natural Resources Canada (2025). Canadian Forest Fire Danger Rating System FWI2025 Update, Section 4.4 "New daily summaries of hourly inputs" and Section 5 "Use and Interpretation."
https://publications.gc.ca/collections/collection_2026/rncan-nrcan/Fo123-2-42-2025-eng.pdf

Github Repository: <https://github.com/lofindanyc/Forest-Fire-Analysis>